

Ticket Sales Analysis & Predictive Modelling — Technical Report

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GitHub Repository: <https://github.com/alexjones602/ticket-sales-analysis>

1. Introduction

When I set up my electronic music event business, I questioned how effective marketing and audience engagement is to ticket sale growth as I am aimed to maximise revenue and reach on my debut event and further events sequentially. This involved me understanding and studying the factors associated with ticket demand to make more informed decisions regarding my event's marketing and promotion.

Primarily, this project analyses ticket sales data collected throughout the timeline of the event from early July up until event day – 21/8/26. Specific data variables include Instagram followers, Reel views, Number of days of advertisement. Analysis was conducted mainly using Jupyter and Python to interpret and identify relationships within the raw dataset and further investigate factors correlated with ticket sales. Additionally, toy datasets were explored for me to gain additional context and knowledge.

Progression of the project follows from initial data cleaning and exploratory analysis to thorough statistical testing, regression analysis, and predictive modelling. Overall, the main aim is to determine whether these analytical techniques can provide meaningful insights into ticket demand and be used to complement evidence-based decision-making.

2. Business Problem

Understanding what induces ticket demand is especially important for event organisers such as myself as resources need to allocate specifically to ensure effective marketing. From a personal experience and perspective, within this industry, a large amount of organisers rush into the marketing decision making process without deep exploration and understanding of factors that contribute to ticket sales. As a result, it is quite common for minimal ticket sales and consequently, events failing.

Thus, I identified the business problem of understanding and predicting ticket demand for my electronic event(s). Analysis involves investigation of variables such as Instagram following and their potential significant relationship with ticket sales. Further investigation determines whether such relationships can be used to inform future marketing campaigns.

Main project questions:

- **AQ1 – Relationship:** What relationships exist between the key variables in the ticket sales dataset?
- **AQ2 — Statistical significance:** Are the most important observed relationships statistically significant?
- **AQ3 — Prediction:** Which variables are most useful for predicting the target outcome?
- **AQ4 — Model performance:** how effectively can the regression models explain or predict the target outcome?

Findings will most likely support myself in identifying which components of marketing require more and less concentration, planning around marketing campaigns, while also creating a more evidence and effective based approach of predicting overall event demand. However, it is important to distinguish that analysis identifies associations and correlations as opposed to causation. In addition, limitations of the dataset must be applied when informing marketing decisions for future events.

3. Dataset

The raw, initial dataset consists of cumulative ticket sales data collected from my Jigsaw Events VOL001 analytics. Such data fuses ticket sales with information regarding marketing, social media, and other variables to investigate key factors associated with demand.

The initial dataset consists of nine event observations and eleven variables. The key dependent variable is ticket sold, whereas remaining variables stand as independent variables that may contribute to the explanation of ticket demand.

Variable	Type	Description
<i>date</i>	Date	Date data collected
<i>tickets_sold</i>	Numerical	Number of tickets sold
<i>instagram_followers</i>	Numerical	Instagram audience size
<i>reel_views</i>	Numerical	Number of social media Reel views
<i>days_advertised</i>	Numerical	Number of days since event was first advertised
<i>day_of_week</i>	Categorical	Day data collected
<i>month</i>	Categorical	Month data collected
<i>student_term_time</i>	Categorical	Whether the data was collected during student term time or a break
<i>weather</i>	Categorical	Weather conditions
<i>rainfall</i>	Numerical	Rainfall recorded when data collected
<i>temperature</i>	Numerical	Temperature when data collected

Numerical variables were used to create descriptive statistics, correlation analysis, and regression modelling, while categorical variables provide additional context for exploring differences between demand.

Tickets sold was selected as the dependent variable as it represents the primary outcome being monitored. Additionally, with a wide range in tickets sold (cumulatively) across the timescale, it provides an opportunity to investigate why demand differed across the event timeline and whether such marketing variables support the differences in ticket sold.

An obvious, key limitation of the dataset is its small sample size of nine observations. Consequently, this limits the ability to use findings to inform larger scale events' marketing campaigns. Furthermore, this dataset only contains a specific context of variables with other factors such as ticket price, artist popularity, promotional spend etc. not being considered when being potentially significant in predicting tickets sold. This implies that results from this analysis should be viewed as exploratory analysis as opposed to definite evidence of factors that contribute to ticket demand levels.

4. Data Cleaning

Before proceeding with statistical analysis and predictions, the raw dataset was inspected and prepared to guarantee variables were valid in format for analysis. Such cleaning process was used with Python and Pandas library.

Firstly, the dataset was inspected to understand structure, variables, and associated data types, the dataset's completeness. Variables were separated into numerical and categorical areas with *tickets_sold* being identified as the dependent variable of the investigation with other variables such as *instagram_followers*, *reel_views* and *days_advertised* being associated with potential predictors and independent variables.

Following variable identification, the dataset was checked for missing values and hence any further treatment required. Additionally, duplicate records were checked to prevent overrepresentation, providing honest and authentic observations and hence analysis.

Data types were examined to confirm that only valid numerical variables could be used in statistical calculations and models, while categorical variables were treated to provide additional relevant context, alongside the *date* variable being retained.

Potential outliers were investigated mainly through descriptive statistics. This applies particularly to the *tickets_sold* variable as values varied quite considerably across the timescale. No data values ('outliers') were removed in the dataset as extreme movements in tickets sold could represent genuine differences in demand rather than errors within the dataset.

Additional variables were treated for further statistical analysis. For example, *day_of_week* was used to classify events into weekday and weekend classes for the upcoming hypothesis test. Sequentially, the dataset was used consistently across the exploratory analysis, statistical testing, and modelling stages.

Following the cleaning process, the dataset was then sufficient for exploratory analysis and valid to conduct the demonstration of statistical and predictive modelling methods. However as previously mentioned, the key limitation of small observation size implies that modelling the results should be conducted and interpreted cautiously.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand patterns, trends and relationships associated in and between variables in the datasets (both raw and toy). This was completed before I intended to conduct statistical and predictive modelling. Descriptive statistics, histograms, boxplots, and correlation analysis were primarily used to examine the variables.

Cumulative ticket sales showed significant variety across the nine observations, ranging from nine to ninety-eight tickets, mean of thirty-four tickets. Average number of Instagram followers was approximately seventy-three, while Reel views averaged approximately one-thousand-four-hundred

and fifty. Correlation analysis provided a positive relationship between cumulative ticket sales and the three main marketing variables:

<i>Variable</i>	<i>Pearson correlation coefficient (r)</i>
Days Advertised	0.896
Instagram Followers	0.802
Reel Views	0.674

This suggests that greater marketing exposure and social media reach were associated with higher cumulative ticket sales in regard to this sample.

Graphical analysis also highlighted considerable volatility in cumulative ticket sales alongside with some high-value observations. Observations were retained as it represented a genuine rise in demand rather than a data-entry error or outlier.

Overall, EDA identified several relationships which created the opportunity for further thorough investigation. These findings provided the foundation for the sequential hypothesis testing and regression analysis, which involved the assessment of statistical significance and test statistics.

6. Statistical Testing

A statistical hypothesis test was completed to assess whether the observed difference in average event views between weekends and weekdays was statistically significant. Data involved was obtained from the Jigsaw Events advertisement dataset.

The hypotheses were defined as:

- H_0 : Average event views are not higher at weekends.
- H_1 : Average event views are higher at weekends.

A significance level of $\alpha = 0.05$ was used.

A one-tailed Welch independent samples t-test was used to compared mean event views between weekend and weekday observations. Welch's t-test was the desired test statistic as it does not assume equal variances of a sample with two groups of observations.

The mean event views were 13.20 for weekends compared with a mean of 11.92 for weekdays. Although the weekend average was higher, the test produced a t-statistic of 0.192 and a p-value of 0.427.

As the p-value is greater than the 5% significance level, the null hypothesis cannot be rejected.

Consequently, there is insufficient statistical evidence to conclude that weekend event views are higher than weekday event views within this dataset / sample. Although a difference was observed, it could be due to variation within the data rather than systemic difference between weekends and weekdays.

This result shows how crucial statistical testing is when interpreting patterns initially identified in exploratory analysis. Higher means do not always indicate a statistically significant relationship. The main limitation of this dataset (small sample of observations) should also be considered yet again when interpreting the result.

7. Regression Modelling

Regression modelling was used to conduct analysis of marketing variables further to the point where they could potentially explain the variation in cumulative ticket sales and identify which variables have the stronger relationships with the depend variable.

A simple linear regression was firstly produced using Instagram followers as the sole predictor. Such model induced a coefficient of determination (R^2) of 0.644, highlighting that approximately 64.4% of the variation in cumulative ticket sales was explained by Instagram followers regarding this dataset / sample.

A multiple linear regression model was then followed. This model used the following independent variables: Instagram followers, Reel views and Days advertised simultaneously as predictors. This allowed for joint assessment rather than separately investigating each individual relationship.

The multiple regression model produced an R^2 of 0.972 and an adjusted R^2 of 0.955. This infers that the predictors are a strong fit to the sample, collectively explaining majority of the variation in cumulative ticket sales. The key coefficients:

<i>Predictor</i>	<i>Coefficient</i>	<i>p-value</i>
Instagram followers	6.979	0.0046
Reel views	-0.186	0.0029
Days advertised	0.556	0.408

Instagram followers had a positive and statistically significant coefficient, where holding other predictors constant, higher Instagram reach correlates to higher cumulative ticket sales.

Reel views had a statistically negative coefficient in the multiple regression model, which was interesting as it had a positive simple correlation with cumulative ticket sales as a sole predictor. As a result, this difference suggests that the relationship between this area of social media engagement and ticket sales is more intricate when other variables are controlled for, inferring interaction or multicollinearity between the variables assessed.

Although days advertised had the strongest simple correlation with cumulative ticket sales ($r = 0.896$), contrastingly, its coefficient was not statistically significant in the multiple regression model ($p = 0.408$). This demonstrates how relationships identified correlation can differ when other predictors are considered simultaneously.

Regression results show that Instagram following may be especially useful when explaining cumulative ticket sales. Moreover, a large R^2 of 0.972 should be interpreted with caution as the number of observations in this sample is only nine alongside a minimal number of predictors (3). As a result, this signifies a risk of overfitting, and such results and conclusions cannot be applied to future events.

Such models provide useful exploratory results about relationships within the dataset but should not be interpreted as concrete evidence of causation or credible forecasting without the implementation of a larger set of data.

8. Model Comparison

A comparison between models was created to assess whether increasing the level of complexity and engineered features within a model resulted in better predictive performance. Consequently, two

Random Forest regression models were conducted using the same train-test split, allowing comparison of outcomes with reliable metrics and ease.

Firstly, the original Random Forest model using Instagram followers, Reel views and Days advertised induced the creation of the improved model which built on the original and contained feature engineered variables with intent to provide more predictive performance.

Both models used a 20% test set and the same random state settings, to allow for a credible comparison with equal training and testing observations.

Such models provided the following to allow for comparisons:

<i>Model</i>	<i>R²</i>	<i>RMSE</i>
Original Random Forest	0.856	7.98
Improved Random Forest	0.848	8.18

R² measures the proportion of variation in cumulative ticket sales explained by the model, with higher values indicating better performance. RMSE measures the typical size of prediction errors, meaning lower values indicate better predictive accuracy.

Surprisingly, the improved Random Forest model did not outperform the original which was unexpected. R² decreased slightly from 0.856 to 0.848, while RMSE rose from 7.98 to 8.18. Consequently, the original model provided stronger performance using the same held dataset.

In addition, this shows that adding features or increasing model complexity does not always improve accuracy in predictions. In this scenario, engineered features when added did not cover for the increased complexity.

However, results as previously mentioned, should be judged cautiously as sample size is small, meaning that 20% of test set contains only two observations. Consequently, performance metrics are sensitive in relation to individual variables and cannot show how model would perform on a larger dataset. Further sets of data would need to be provided before applying to real-world forecasting.

9. Key Findings

Analysis produced three main findings related to marketing activity, social media engagement, and cumulative ticket sales.

Finding 1 – Days advertised strong relationship with cumulative ticket sales.

Evidence: Days advertised had the strongest correlation coefficient with cumulative ticket sales ($r = 0.896$).

Implication: As time increased from the day since the event was first advertised, ticket sales have as a result increased. However, as ticket sales are cumulative in this dataset, this relationship is partially expected where a longer marketing campaign allows for an accumulation of ticket sales. Consequently, this relationship should be interpreted as an association rather than evidence that advertising duration provides additional sales directly.

Finding 2 – Weekend event views were not significantly greater than weekday views.

Evidence: Mean event views were larger at weekends than on weekdays. However, the Welch one tailed t-test produced a test static of 0.192 and a p-value of 0.42, which is above the 0.05 significance threshold.

Implication: Hence there is insufficient evidence to say that weekend marketing provided higher event views. Moreover, marketing decisions should not assume that weekend activity is more optimal and effective.

Finding 3 – Instagram following size as a key predictor of cumulative ticket sales.

Evidence: Instagram following as a variable had a positive regression coefficient of 6.979 with a p-value of 0.0046, highlighting a statistically significant positive correlation with cumulative ticket sales when predictors are held constant.

Implication: Strong social media presence may be a key factor to consider when promoting a future event. However, the result does not imply direct causation.

Overall, analysis shows that marketing exposure and social media presence are particularly strong in association with cumulated tickets being sold within this dataset. Days advertised as a variable showed the strongest simple relationship, while Instagram followers highlighted a statistically significant predictor in the multiple regression model.

10. Business Implications

Such findings provide valuable factors that should be considered and learnt from for future business opportunities / events.

The strong relationship between days advertised and cumulative ticket sales ($r = 0.896$), suggests that campaign timing is a major component for planning events. Starting promotion earlier than anticipated may allow for ticket sales to accumulate even more, however this is not for definite.

Instagram followers were also identified as a statistically significant positive predictor of cumulative ticket sales. Suggesting that prioritising a strong and active social media platform could support ticket demand.

The hypothesis test found no sufficient evidence that there is a credible difference between weekend and weekday event views. Therefore, marketing campaigns should not be allocated solely to weekend days without further evidence and investigation.

11. Limitations

Most importantly, the small sample size of nine observations limits both the reliability and generalisability of the findings, affecting its suitability for future events.

The cumulative nature of the tickets sold variable also affects the analysis, where longer marketing campaigns naturally provide more time for tickets to accumulate.

Additionally, the dataset does not include several other important predictors and variables such as ticket price, artist reach, advertising expenditure etc.

Finally, the data used is simply observational where the relationships explored cannot be identified as evidence of causation.

12. Future Improvements

Firstly, the most crucial improvement would be to collect data from a larger number of dates throughout the campaign, providing a more statistical and thorough analysis approach while also increasing reliability.

Additional variables should be collected in the dataset such as ticket price, advertising expenditure, and website visits. Advanced engineered variables should be created as well, for example tickets sold per day or sales velocity as opposed to cumulative ticket sales. This provides a clearer insight into how fast demand is developing over time.

Moreover, with more data available alongside more differentiated techniques such as cross-validation, hyperparameter tuning and further machine-learning models, in aggregate can be used to improve the forecasting process and reliability of models.

13. Conclusion

This project investigated the relationship marketing reach, social media presence, and cumulative ticket sales for my debut electronic music event. Analysis identified strong linear relationships between ticket sales and both days advertised ($r = 9.896$) and Instagram followers ($r = 0.802$). The Instagram followers variable was also identified as a statistically significant positive predictor in the regression model. However, the small sample size limits the findings' credibility, where it should be considered as exploratory as opposed to definitive.

Overall, the project demonstrates how data analysis, statistical testing and predictive modelling can be used to generate evidence-based interpretations for event marketing. Hopefully with a larger dataset, the approach could be developed into a more reliable and credible ticket-demand forecasting system.